Analytical Chemistry Cumulative Exam
March 8th, 2024
Topic: Key equations in analytical chemistry

Background: Throughout the field of analytical chemistry there are several equations that are fundamental to the basis of the techniques we use. These are often included in talks and papers, in which the presenter will often assume that you know what relationship the equation describes and what the variables in the equation represent. Familiarity with these equations will help you follow scientific discussions, seminars, and read the literature.

Below is a list of key equations from across the field of analytical chemistry. For each equation

i) Provide the name for the equation (if there is one). If there is no common name state 'no common name'. *If multiple names exist, you need only provide one answer.*
ii) Write what each of the variable terms in the equation represents and/or its name (if it has one) (1 sentence or less)
iii) Write what the equation is useful for/when it is applicable. (Note, for some equations this will just be a rewording of the relationship, while in others it might require you to state in what situation the equation is valid). *Expected length is ~1 sentence.*

You are graded on demonstrating your recognition of the equations and understanding what general area they are useful for, not writing details of every specific sub-application, spelling, or grammar. Please keep your answers brief and to the point – nothing beyond the 2nd sentence of a description will be graded.

<u>Example</u>

As an example, if you were given the equation $C = \frac{m}{V \cdot MW}$

A sufficient answer would be: i) No name, ii) C: concentration of a solution, m: mass of analyte, V: volume of solution, MW: molecular weight of analyte. iii) Determining the molar concentration of a solution.

<u>Equations</u>

a) $A = -\log T$
b) $H = A + \frac{B}{u_x} + Cu_x$
c) $\Delta G = -nFE$
d) $t = \frac{L^2}{2D}$
e) $s = \sqrt{\sum_i \frac{(x_i - \bar{x})^2}{n-1}}$
f) $\mu = \frac{1}{2}\sum_i z_i^2 c_i$
g) $N = \frac{16L^2}{w^2}$
h) $t_{\text{calc}} = \left|\frac{\bar{x} - \mu_0}{s/\sqrt{N}}\right|$
i) $E = \frac{hc}{\lambda}$